



Bluetooth flawed for decades

A really old flaw in Bluetooth technologies allows snoopers to steal the data being transferred between devices. <https://dgit.in/Aug18-41-1>



Japan's calendar is broken

Japanese developers have no character to represent the next era after Emperor Akihito steps down. <https://dgit.in/Aug18-41-2>

HOW WE TESTED

The first step while picking the controllers was to select the ones which had the basic buttons and controls. This included the D-pad, analog sticks, triggers and bumpers, and four action buttons. After the selection process, we verified whether they could be detected over Bluetooth and successfully establish a connection. Once paired, if a mobile game has been developed with support for gamepads, it automatically detects the gamepad. Not all games give you a pop-up alert when a gamepad is connected. In cases where mobile games don't support gamepads, third-party emulators can be used to assign and bind different actions to the buttons. However, officially supported games will always offer the best experience. We wanted to test the gamepads based on what they offered out of the box and how the manufacturer wants you to experience the product. If they have developed their own emulator app, we used it.

Picking the games and devices

Coming to the choice of games, it's always best to consider officially supported games listed on the gamepad manufacturer's website. Gamepads are best suited to be used in side-scrolling, racing and shooting games. Except one, none of the controllers supported MOBA games officially. Hence, we didn't cover them in the comparison. We tried to pick common games that were supported by all the gamepads included in our comparison. They are as follows:

- Asphalt 8
- Modern Combat 5
- Sonic Classic

To keep the testing method standard, we used all the gamepads on two devices, including a OnePlus 5T for Android compatible gamepads, and a 12.9-inch iPad Pro for iOS games. In this comparison, the GameSir G5 and SteelSeries Stratus were the only ones

compatible with iOS with the latter being compatible only with iOS.

Performance testing

We didn't have access to any benchmarking software or hardware tools to objectively test the performance of gamepads. Hence, most of the testing was based on a subjective point of view. After the gamepads were connected to the devices, the selected games were played. In those games, the same level or sequence was replayed with each gamepad. This ensured consistency, giving us a standardised scenario to test each gamepad. We observed responsiveness and latency. Our goal was to check whether the level of responsiveness was different on any of these gamepads. Any kind of weird behaviour we encountered while playing would result in a lowered score for that gamepad. Tests were run multiple times to ensure the errors weren't a one-time anomaly. The ease of pairing the gamepad with the devices was also given some weight. Any extra steps required by the gamepad, other than the regular Bluetooth pairing, was given a lower score.

Features

Although the basic features were covered by all the gamepads, a couple of them offered something extra. Whether it was better compatibility or extra connectivity options, they were scored accordingly. If the gamepad offered extra buttons apart from the action buttons and triggers, they were given a higher score. In terms of software or app support, if the manufacturer offered their own emulator that enhanced the functionality of the gamepad, they were given better scores. However, this also meant that it required extra steps before playing the games. So, it lost points on performance testing.

Build quality

As a gamepad, it's naturally going to be under a lot of stress. The buttons will be subject to

a lot of abuse, especially while playing platformer or shooting games. Since we don't have the right equipment to determine whether the buttons are objectively built well, we haven't scored them. However, tactile and submarine microswitches have better durability than membrane switches. Membrane switches are pretty common and found on the D-pad, action buttons, triggers and bumpers. If the controller employed the use of tactile micro-switches, they were rated higher. For the exterior, we scored based on the coating over the body and palm grip areas. The upper layer coating and choice of material over the joysticks was also scored.

Verdict

After testing all the gamepads following the above methodology, we give them scores based on these parameters including an overall score. These scores determine the winner in three categories. The first is the Best Performer which will be won by the gamepad that delivers exceptional performance and an overall superiority over the rest of the gamepads. Essentially, it will score the highest points in our testing and it will automatically become our most recommended gamepad.

The second award is the Best Buy. If you're offering amazing performance and features, it's not always going to be affordable. Hence, this award will ensure that the gamepad offering the best value for money is given some recognition. It won't necessarily be the most affordable one but it will surely offer the best balance between its overall performance and price.

Our final award is the Editor's Pick. This particular award will be given to the gamepad which the editor found to be the best during the comparison test. It's completely subjective and it's possible that it could have won an award from the previous two categories.

you get used to playing games on your smartphones. However, what are you going to do about those cramped up fingers? Holding your smartphone like a controller isn't really the most ergonomic thing. After about an hour, your hand will start aching, forcing you to take a break, and if you persist, you risk potentially causing permanent harm to your hand in the future. Since your screen is your controller, you're

bound to hold it in front of your eyes, causing even more discomfort to your hands. The natural solution in these times is to use a controller or gamepad. Not only are they ergonomic, but gaming becomes much easier with them. You don't have to struggle anymore while tapping on different buttons on that small screen. With a gamepad, your smartphone's display will function only as a display henceforth,

taking away any form of obstruction when you try to tap on the screen.

Currently, the market is flooded with mobile gamepads. Flipkart and Amazon throw hundreds of search results across a wide price range. If you keep looking, you'll find similar looking gamepads with different names. That's because those particular brands or manufacturers are OEM devices which are simply rebranded